

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihalan Produk:

Product Description:

Cat No. :

Synonyms

CAS No

Molecular Formula

**Acetic anhydride**
**Acetic anhydride**

A/0440/PB08, A/0440/PB17

Acetyl oxide, Acetic acid anhydride, Acetic oxide, Ethanoic anhydride

108-24-7

C4 H6 O3

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use

Laboratory chemicals.

Uses advised against

No Information available

**Company**

 Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
 Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,  
 No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,  
 Selangor Darul Ehsan, Malaysia.  
 Main line: +60 3-5525 7888

**Supplier**

E-mail address

Enquiry.my@thermofisher.com

**Emergency Telephone Number**

Tel: +03-5525 7888

 CHEMTREC Malaysia **1-800-815-308** (Malay)

 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                    |                     |
|------------------------------------|---------------------|
| Flammable liquids                  | Category 3 (H226)   |
| Acute oral toxicity                | Category 4 (H302)   |
| Acute Inhalation Toxicity - Vapors | Category 2 (H330)   |
| Skin Corrosion/Irritation          | Category 1 B (H314) |
| Serious Eye Damage/Eye Irritation  | Category 1 (H318)   |

**Label Elements**


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## Signal Word

## Danger

### Hazard Statements

H226 - Flammable liquid and vapor  
H302 - Harmful if swallowed  
H314 - Causes severe skin burns and eye damage  
H330 - Fatal if inhaled

### Precautionary Statements

#### Prevention

P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P240 - Ground and bond container and receiving equipment  
P241 - Use explosion-proof electrical/ ventilating/ lighting equipment  
P242 - Use non-sparking tools  
P243 - Take action to prevent static discharges  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P284 - Wear respiratory protection

#### Response

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P310 - Immediately call a POISON CENTER or doctor  
P330 - Rinse mouth  
P331 - Do NOT induce vomiting  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P362 + P364 - Take off contaminated clothing and wash it before reuse

#### Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
P405 - Store locked up

#### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

### Other Hazards

EUH071 - Corrosive to the respiratory tract  
Lachrymator (substance which increases the flow of tears)  
Reacts with water and forms acetic acid  
This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component        | CAS No   | Weight % |
|------------------|----------|----------|
| Acetic anhydride | 108-24-7 | >99      |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical

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attention is required.

## Ingestion

Do NOT induce vomiting. Call a physician or poison control center immediately.

## Inhalation

Remove to fresh air. If breathing is difficult, give oxygen. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

## Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

## Most important symptoms and effects, both acute and delayed

Causes burns by all exposure routes. Difficulty in breathing. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.

## Indication of any immediate medical attention and special treatment needed

### Notes to Physician

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### Suitable Extinguishing Media

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers.

#### Extinguishing media which must not be used for safety reasons

Water.

### Special hazards arising from the substance or mixture

Flammable. Corrosive material. Water reactive. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Containers may explode when heated. Thermal decomposition can lead to release of irritating gases and vapors.

### Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Use personal protective equipment as required. Evacuate personnel to safe areas. Remove all sources of ignition. Take precautionary measures against static discharges. Avoid contact with skin, eyes and inhalation of vapors.

### Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

### Methods and Material for Containment and Cleaning Up

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Remove all sources of ignition. Do not expose spill to water. Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Use spark-proof tools and explosion-proof equipment.

## Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Use only under a chemical fume hood. Wear personal protective equipment/face protection. Keep away from open flames, hot surfaces and sources of ignition. Use spark-proof tools and explosion-proof equipment. Do not breathe mist/vapors/spray. Do not get in eyes, on skin, or on clothing. Do not ingest. If swallowed then seek immediate medical assistance. Do not allow contact with water.

### Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. Keep away from water or moist air. Flammables area.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component        | Malaysia | ACGIH TLV                 | OSHA PEL                                                                                                       |
|------------------|----------|---------------------------|----------------------------------------------------------------------------------------------------------------|
| Acetic anhydride |          | TWA: 1 ppm<br>STEL: 3 ppm | (Vacated) Ceiling: 5 ppm<br>(Vacated) Ceiling: 20 mg/m <sup>3</sup><br>TWA: 5 ppm<br>TWA: 20 mg/m <sup>3</sup> |

  

| Component        | European Union | The United Kingdom                                                                                              | Germany                                                                                                                                                                                                                                                           |
|------------------|----------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acetic anhydride |                | STEL: 2 ppm 15 min<br>STEL: 10 mg/m <sup>3</sup> 15 min<br>TWA: 0.5 ppm 8 hr<br>TWA: 2.5 mg/m <sup>3</sup> 8 hr | TWA: 0.1 ppm (8 Stunden). AGW - exposure factor 2<br>TWA: 0.42 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 0.1 ppm (8 Stunden). MAK<br>TWA: 0.42 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 0.2 ppm<br>Höhepunkt: 0.84 mg/m <sup>3</sup> |

### Exposure Controls

#### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

Eye Protection

Goggles

Hand Protection

Protective gloves

Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure

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Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.  
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

## Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

## Recommended Filter type:

Organic gases and vapours filter Type A Brown conforming to EN14387

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

## Environmental exposure controls

No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

**Appearance** Colorless

**Physical State** Liquid

**Odor** pungent

**Odor Threshold** No data available

**pH** 3

**Melting Point/Range** -73.1 °C / -99.6 °F

**Softening Point** No data available

**Boiling Point/Range** 140 °C / 284 °F

**Flash Point** 49 °C / 120.2 °F

@ 760 mmHg

**Method** - CC (closed cup)

**Evaporation Rate** 0.46

(Butyl Acetate = 1.0)

**Flammability (solid,gas)** Not applicable

Liquid

**Explosion Limits** **Lower** 2 Vol%

**Upper** 12 Vol%

**Vapor Pressure** 5 mbar @ 20 °C

**Vapor Density** 3.5

(Air = 1.0)

**Specific Gravity / Density** 1.087

**Bulk Density** Not applicable

Liquid

**Water Solubility** hydrolyses

**Solubility in other solvents** No information available

### Partition Coefficient (n-octanol/water)

**Component** **log Pow**

Acetic anhydride -0.27

**Autoignition Temperature** 316 °C / 600.8 °F

**Decomposition Temperature** No data available

**Viscosity** 0.91 mPa.s at 20 °C

**Explosive Properties**

explosive air/vapour mixtures possible

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**Oxidizing Properties** No information available

**Molecular Formula** C4 H6 O3  
**Molecular Weight** 102.09

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

Yes.

### Chemical Stability

Moisture sensitive.

### Possibility of Hazardous Reactions

**Hazardous Polymerization** Hazardous polymerization does not occur.  
**Hazardous Reactions** No information available.

### Conditions to Avoid

Incompatible products. Keep away from open flames, hot surfaces and sources of ignition.  
Exposure to moist air or water.

### Incompatible Materials

Strong acids. Water. Strong reducing agents. Alcohols. Bases. Oxidizing agent.

### Hazardous Decomposition Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Product Information

#### (a) acute toxicity;

**Oral** Category 4  
**Dermal** Based on available data, the classification criteria are not met  
**Inhalation** Category 2

| Component        | LD50 Oral                                 | LD50 Dermal                  | LC50 Inhalation                                                        |
|------------------|-------------------------------------------|------------------------------|------------------------------------------------------------------------|
| Acetic anhydride | LD50 = 630 mg/kg (Rat)<br>Equiv. OECD 410 | LD50 = 4000 mg/kg ( Rabbit ) | LC100: 1.67 mg/L/6h (Rat)<br>Equiv. OECD 412<br>LC50: 400 ppm/6h (Rat) |

#### (b) skin corrosion/irritation;

Category 1 B  
Based on available literature and data from closely analogous substances using  
structure/activity relationships

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|                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (c) serious eye damage/irritation;                                                                                           | Category 1<br>Based on available literature and data from closely analogous substances using structure/activity relationships                                                                                                                                                                                                                    |
| (d) respiratory or skin sensitization;<br>Respiratory<br>Skin                                                                | Based on available data, the classification criteria are not met<br>Based on available data, the classification criteria are not met                                                                                                                                                                                                             |
| (e) germ cell mutagenicity;                                                                                                  | Based on available data, the classification criteria are not met<br>Not mutagenic in AMES Test                                                                                                                                                                                                                                                   |
| (f) carcinogenicity;                                                                                                         | Based on available data, the classification criteria are not met<br>There are no known carcinogenic chemicals in this product                                                                                                                                                                                                                    |
| (g) reproductive toxicity;                                                                                                   | Based on available data, the classification criteria are not met                                                                                                                                                                                                                                                                                 |
| (h) STOT-single exposure;                                                                                                    | Based on available data, the classification criteria are not met                                                                                                                                                                                                                                                                                 |
| Study/Evidence (Classification basis)<br>Test species / Sex / Route of exposure<br>Effective dose<br>Results / Target organs | OECD Test No. 412<br>Rat / Inhalation (4h)<br>LOAEL = 300 - 2000 ppm<br>Eyes.                                                                                                                                                                                                                                                                    |
| (i) STOT-repeated exposure;                                                                                                  | Based on available data, the classification criteria are not met                                                                                                                                                                                                                                                                                 |
| Test method<br>Test species / Duration<br>Study result<br>Route of exposure<br>Target Organs                                 | OECD Test No. 413<br>Rat / 90 days<br>NOAEL = 0.2 - 1 ppm<br>Inhalation<br>None known.                                                                                                                                                                                                                                                           |
| (j) aspiration hazard;                                                                                                       | Based on available data, the classification criteria are not met                                                                                                                                                                                                                                                                                 |
| Symptoms / effects, both acute and delayed                                                                                   | Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. |
| Endocrine Disrupting Properties                                                                                              | Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.                                                                                                                                                                                                              |

## SECTION 12: ECOLOGICAL INFORMATION

|                                                                      |                                                                                                                                                             |
|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <u>Ecotoxicity effects</u>                                           | Reacts with water so no ecotoxicity data for the substance is available. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. |
| <u>Persistence and degradability</u><br>Persistence<br>Degradability | Readily biodegradable<br>Persistence is unlikely, based on information available.<br>Reacts with water.                                                     |

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## Degradation in sewage treatment plant

Decomposes in contact with water. Neutralization is normally necessary before waste water is discharged into water treatment plants.

## Bioaccumulative potential

Bioaccumulation is unlikely

| Component        | log Pow | Bioconcentration factor (BCF) |
|------------------|---------|-------------------------------|
| Acetic anhydride | -0.27   | 3.16                          |

## Mobility in soil

Decomposes in contact with water. .

## Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

## Other adverse effects

No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

#### Waste from Residues/Unused Products

Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

#### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point.

#### Other Information

Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains Do not flush to sewer Large amounts will affect pH and harm aquatic organisms

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

UN-No UN1715  
Hazard Class 8  
Subsidiary Hazard Class 3  
Packing Group II  
Proper Shipping Name ACETIC ANHYDRIDE

### Road and Rail Transport

UN-No UN1715  
Hazard Class 8  
Subsidiary Hazard Class 3  
Packing Group II  
Proper Shipping Name ACETIC ANHYDRIDE

### IATA

UN-No UN1715  
Hazard Class 8  
Subsidiary Hazard Class 3  
Packing Group II  
Proper Shipping Name ACETIC ANHYDRIDE

## Special Precautions for User

No special precautions required

## SECTION 15: REGULATORY INFORMATION

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## Safety, health and environmental regulations/legislation specific for the substance or mixture

### International Inventories

X = listed

| Component        | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Acetic anhydride | 203-564-8 | X    | X   | X     | X    | X    | X     | X    | KE-00017 |

### National Regulations

#### Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance

This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### **Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Revision Date**

23-Mar-2025

**Revision Summary**

SDS sections updated, 1, 3, 8, 11, 15.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

### **Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the

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date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**